

SQL Data Extraction & Analysis Report

Phase: Backend Data Warehousing & Logic Building

Database: MySQL

Overview:

Before visualizing the data in Power BI, raw financial data was loaded into a MySQL relational database. Complex SQL queries were executed to clean, aggregate, and validate the business logic. Below are the core business questions solved using SQL, along with their respective queries and output summaries.

Question 1: Which countries are generating the highest gross sales, and what is their actual net profit?

Business Logic: To identify if high sales volume directly translates to high profitability across different regions.

SQL Query:

```
SELECT Country, ROUND(SUM(Sales), 2) AS Total_Sales, ROUND(SUM(Profit), 2) AS Net_Profit
FROM financial_sales_data
GROUP BY Country
ORDER BY Total_Sales DESC;
```

Output Summary:

The query successfully aggregated millions of rows, revealing that while certain countries like the USA and Canada lead in gross sales revenue, their net profit margins are comparatively lower due to higher operational costs or excessive discounting.

Question 2: What is the revenue contribution of each business segment?

Business Logic: To understand our customer base and see which sector (Government, Midmarket, Small Business, etc.) brings in the most orders and revenue.

SQL Query:

```
SELECT Segment, COUNT(*) AS Total_Transactions, ROUND(SUM(Sales), 2) AS Total_Revenue
FROM financial_sales_data
GROUP BY Segment
ORDER BY Total_Revenue DESC;
```

Output Summary:

The output confirmed a heavy reliance on the 'Government' and 'Midmarket' segments, which together contribute to over 60% of the total revenue, while 'Small Business' remains an under-penetrated market.

Question 3: How do different Discount Bands (High, Medium, Low, None) affect the profitability of our top products?

Business Logic: To evaluate if giving high discounts on hero products like 'Paseo' and 'Velo' is actually profitable or just inflating raw sales numbers.

SQL Query:

```
SELECT Product, Discount_Band, ROUND(SUM(Sales), 2) AS Gross_Sales,  
ROUND(SUM(Profit), 2) AS Actual_Profit  
FROM financial_sales_data  
GROUP BY Product, Discount_Band  
ORDER BY Actual_Profit DESC;
```

Output Summary:

This query provided a critical insight: Our top-selling products often yield the lowest net profit when sold under the 'High' discount band. It proved that selling fewer units at a 'Low' or 'None' discount band is more profitable for the company.

Question 4: What is the month-over-month (MoM) profit trend for the fiscal years 2013 and 2014?

Business Logic: To map out seasonal spikes in profitability and identify the strongest financial quarters.

SQL Query:

```
SELECT Year, Month_Name, ROUND(SUM(Profit), 2) AS Monthly_Profit  
FROM financial_sales_data  
GROUP BY Year, Month_Name, Month_Number  
ORDER BY Year ASC, Month_Number ASC;
```

Output Summary:

The chronological extraction showed extreme volatility in profit trends, with significant peaks consistently appearing at the end of fiscal quarters (e.g., October, December), indicating aggressive end-of-quarter sales pushes.

Question 5: What is the actual Profit Margin Percentage for each product?

Business Logic: High profit in numbers doesn't always mean high efficiency. We need to calculate the margin percentage (Profit divided by Sales) to see which product gives the best return on investment.

SQL Query:

```
SELECT Product, ROUND(SUM(Sales), 2) AS Total_Sales, ROUND(SUM(Profit), 2) AS  
Total_Profit, ROUND((SUM(Profit) / SUM(Sales)) * 100, 2) AS Profit_Margin_Percent  
FROM financial_sales_data  
GROUP BY Product  
ORDER BY Profit_Margin_Percent DESC;
```

Output Summary:

This advanced calculation revealed the true efficiency of our catalog. While 'Paseo' had the highest total sales, products like 'Amarilla' or 'VTT' actually showed a higher profit margin percentage, meaning they cost less to sell and yield better proportional returns.

Question 6: Who are the Top 3 performing countries within each business Segment?

Business Logic: To find regional leaders inside specific sectors (e.g., who is the top buyer in the Government sector vs. the Midmarket sector) using advanced SQL Window Functions.

SQL Query:

```
WITH RankedSales AS (  
SELECT Segment, Country, ROUND(SUM(Sales), 2) AS Segment_Sales, DENSE_RANK()  
OVER (PARTITION BY Segment ORDER BY SUM(Sales) DESC) as Sales_Rank  
FROM financial_sales_data  
GROUP BY Segment, Country  
)  
SELECT Segment, Country, Segment_Sales, Sales_Rank  
FROM RankedSales  
WHERE Sales_Rank <= 3;
```

Output Summary:

Using the DENSE_RANK() window function, we successfully partitioned the data. The output provided the CFO with a clear top 3 ranking of countries for every single segment, making it easier to target marketing budgets regionally based on segment strength.

Question 7: What is the Cumulative (Running Total) of Profit over the months in the year 2014?

Business Logic: To track how our total profit builds up continuously over the year, helping the finance team see if we are hitting our annual financial targets month-by-month.

SQL Query:

```
WITH MonthlyData AS (  
SELECT Month_Number, Month_Name, ROUND(SUM(Profit), 2) AS Monthly_Profit
```

```

FROM financial_sales_data

WHERE Year = 2014

GROUP BY Month_Number, Month_Name

)

SELECT Month_Name, Monthly_Profit, ROUND(SUM(Monthly_Profit) OVER (ORDER
BY Month_Number), 2) AS Running_Total_Profit

FROM MonthlyData;

```

Output Summary:

By applying a rolling SUM over the ordered months, the query generated a running total of profits. This allowed us to map the precise trajectory of our 2014 financial growth, confirming that the largest jump in cumulative profit occurred between Q3 and Q4.

Question 8: What is the total number of 'Units Sold' for each Product, and which product is moving the fastest?

Business Logic: Profit is important, but inventory movement (velocity) is equally crucial. We need to know which product is leaving the warehouse the fastest, regardless of its price.

SQL Query:

```

SELECT Product, ROUND(SUM(Units_Sold), 2) AS Total_Units_Sold

FROM financial_sales_data

GROUP BY Product

ORDER BY Total_Units_Sold DESC;

```

Output Summary:

The extraction revealed that 'Paseo' and 'VTT' are the highest volume drivers in terms of sheer units moved. This tells the supply chain team to keep the warehouse fully stocked with these specific products to avoid out-of-stock situations.

Question 9: Are there any transactions where the company suffered a net loss (Negative Profit)?

Business Logic: To identify if our discounting strategy or operational costs ever caused us to lose money on a sale, which is a critical red flag for the CFO.

SQL Query:

```

SELECT Country, Segment, Product, Discount_Band, Sales, Profit

FROM financial_sales_data

WHERE Profit < 0

ORDER BY Profit ASC;

```

Output Summary:

By filtering for negative profit, the query successfully isolated specific transactions where the company lost money. It showed that excessive 'High' discount bands in certain segments directly resulted in negative margins, highlighting areas where discounting rules must be strictly revised.

Question 10: What is the Average Discount Amount given across different Business Segments?

Business Logic: To see which sector negotiates the hardest and demands the highest average discounts from our sales team.

SQL Query:

```
SELECT Segment, ROUND(AVG(Discounts), 2) AS Average_Discount
FROM financial_sales_data
GROUP BY Segment
ORDER BY Average_Discount DESC;
```

Output Summary:

The data showed that the 'Enterprise' and 'Government' segments receive the highest average discounts per transaction. This makes sense for bulk orders, but requires close monitoring to ensure salespeople aren't giving away too much margin just to close deals.

Question 11: How does the Total Cost of Goods Sold (COGS) compare to Gross Sales for each Country?

Business Logic: Sales revenue doesn't mean much if the cost to manufacture and deliver the goods (COGS) is too high. This query compares the two to check gross efficiency by region.

SQL Query:

```
SELECT Country, ROUND(SUM(Sales), 2) AS Total_Gross_Sales, ROUND(SUM(COGS),
2) AS Total_COGS, ROUND(SUM(Sales) - SUM(COGS), 2) AS Gross_Margin
FROM financial_sales_data
GROUP BY Country
ORDER BY Gross_Margin DESC;
```

Output Summary:

This comparative aggregation provided a clear picture of operational efficiency. Countries like Germany and France showed a very healthy gap between Sales and COGS, proving them to be highly efficient and profitable operational hubs.

Question 12: Which are the Top 3 most profitable products specifically within the 'Government' segment?

Business Logic: Since 'Government' is our biggest segment, we need a targeted analysis to find out exactly which products they prefer, enabling highly targeted future marketing.

SQL Query:

```
SELECT Product, ROUND(SUM(Profit), 2) AS Gov_Total_Profit
FROM financial_sales_data
WHERE Segment = 'Government'
GROUP BY Product
ORDER BY Gov_Total_Profit DESC
LIMIT 3;
```

Output Summary:

Filtering strictly by the Government segment, the query ranked the top 3 products. 'Paseo' and 'Velo' emerged as the absolute favorites. This insight tells the product team to prioritize public sector compliance for these specific items.

Question 13: What is the Year-over-Year (YoY) Sales Comparison for each month between 2013 and 2014?

Business Logic: To evaluate if the business is growing or shrinking compared to the exact same time last year.

SQL Query:

```
SELECT Month_Name,
SUM(CASE WHEN Year = 2013 THEN Sales ELSE 0 END) AS Sales_2013,
SUM(CASE WHEN Year = 2014 THEN Sales ELSE 0 END) AS Sales_2014
FROM financial_sales_data
GROUP BY Month_Name, Month_Number
ORDER BY Month_Number ASC;
```

Output Summary:

Using conditional aggregations (CASE statements), this advanced query pivoted the data to compare 2013 and 2014 side-by-side. It proved that late Q3 and Q4 of 2014 saw massive YoY growth compared to 2013, confirming that our recent sales strategies are highly effective.

Question 14: What is the overall percentage contribution of each Country to the Global Total Profit?

Business Logic: To understand the global power dynamics of our business. If one country contributes 40% of all profits, a recession there could crash the whole company.

SQL Query:

```
WITH TotalGlobal AS (  
SELECT SUM(Profit) AS Global_Profit FROM financial_sales_data  
)  
SELECT Country, ROUND(SUM(Profit), 2) AS Country_Profit, ROUND((SUM(Profit) /  
(SELECT Global_Profit FROM TotalGlobal)) * 100, 2) AS Contribution_Percentage  
FROM financial_sales_data  
GROUP BY Country  
ORDER BY Contribution_Percentage DESC;
```

Output Summary:

This complex query utilized a CTE to first calculate the global baseline, then calculated each country's proportional slice. The CFO can now clearly see that North America (USA & Canada) drives the lion's share of global profits, identifying both a strength and a potential geographic risk.